

CONDITIONAL VALUE OF PERMUTATION SCORES IN FROZEN MULTILINGUAL ASR RERANKING

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ABSTRACT

We assess permutation scoring on shared ASR candidate banks: three frozen Whisper-small checkpoints, five languages, one main fit per selector and 30-second inputs. Global adds a permutation contrast to clean-audio confidence, length and consensus (Conf). Relative to Conf, it lowers macro word error rate (WER) by 0.1410 percentage points on FLEURS and 0.1824 on Common Voice v27 (CV27). Frozen masked-language-model scoring (Text) improves on Global by 0.9864 and 1.2043 points. Adding Global to Text changes CV27 WER by -0.0104 points (adjusted 98.33% client interval $[-0.0280, +0.0069]$), without establishing further gain. Post-exposure controls do not establish a contrast-specific advantage; TRAIN-subset refits show sensitivity. The original Local-over-Global increment is unestablished; minimum-WER fitting likewise does not establish gains over corresponding ridge fits. Measured costs include score acquisition and Text’s additional resources. Evidence supports modest gains beyond internal confidence, with further benefit beyond external text unestablished.

Index Terms—speech recognition, reranking, perturbation features, multilingual evaluation, candidate selection

1. INTRODUCTION

An improvement over beam search alone cannot isolate the value of a perturbation-derived ASR score. Clean-audio confidence [1], length and candidate consensus already inform selection; external text provides a further, resource-intensive control. The relevant question is whether scoring disrupted audio adds predictive value after these signals are available.

We test this question on shared frozen candidate banks. Matched feature ablations isolate the added score from candidate quality; duplicate-coordinate and alternative-parameterization controls test simpler explanations for ridge gains. Inference measurements include candidate and score acquisition.

Across five languages and three recognizer checkpoints, permutation scores yield small gains over internal confidence, without established gains beyond external text. Our contribution is a controlled assessment of scoring information, parameterization, fit sensitivity and resources.

2. RELATED WORK

Adaptation and reranking. LoRA-Whisper [2] and language experts [3] adapt ASR; we freeze it. Multilingual rescoring combines acoustics and text [4, 5]; unlike [5], we assume known language. RescoreBERT [6] trains a BERT reranker; we use established pseudo-log-likelihood (PLL) scoring [7] from a frozen masked language model (MLM).

Scoring and objectives. MBR [8], minimum-WER training [9] and speech/text rescoring [10] provide precedents. Whisper-CD [11] and LCAR [12] alter token-wise decoding; our candidate-level CD-seq does not reproduce them. Token-probability [13] and normalization [14] analyses complement our feature comparisons.

3. CANDIDATE SCORING AND SELECTION

3.1. Candidates and perturbation response

Frozen Whisper-small [15] with rank-four LoRA [16] (Fig. 1) produces greedy (G) and five-best outputs; B5 is the beam winner b . WER-v3 normalization deduplicates them in beam-then-greedy order. Banks contain at most six candidates; singletons return beam.

For log-mel input X , a recording-hash permutation T reorders 20-frame blocks (about 200 ms), including the final shorter block, before the encoder; padding and frame count stay fixed. Identity permutations rotate if two or more blocks exist. Candidate prefixes supply teacher-forced clean/permutated body-token log probabilities $\ell_t(y, X)$ and $\ell_t(y, T(X))$. Decoder inputs shift once. Special tokens and end-of-sequence (EOS) are excluded; capped outputs receive no inserted EOS. Empty means are zero. For clean/permutated body means $c(y), q(y)$,

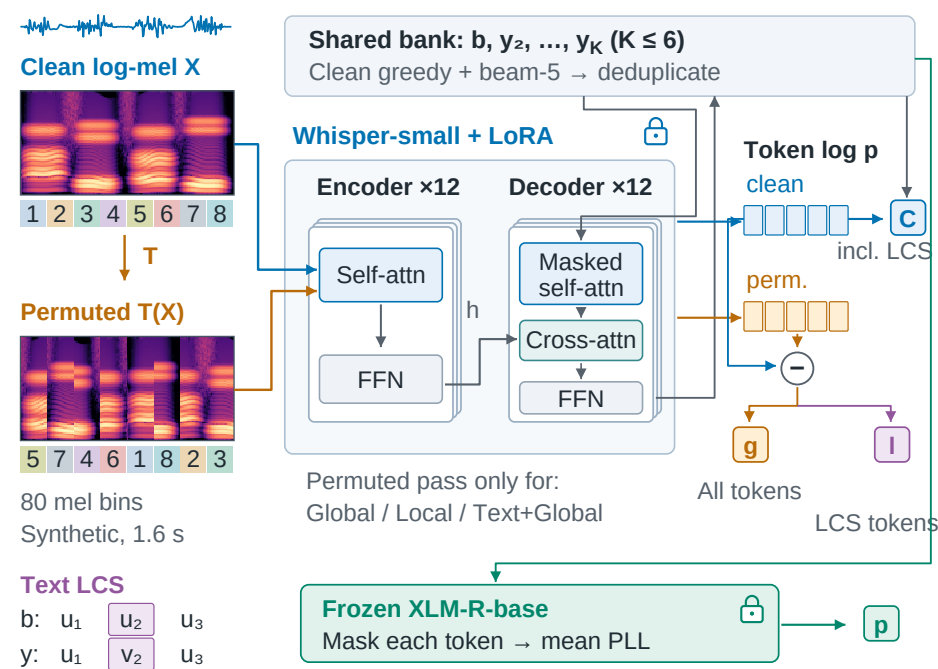
$$\begin{aligned} a_t(y) &= \ell_t(y, X) - \ell_t(y, T(X)), \\ A_g(y) &= \text{mean}_{t \in \text{body}(y)} a_t(y) = c(y) - q(y). \end{aligned} \quad (1)$$

Local tests whether focusing the score on candidate disagreements improves selection. A longest common subsequence (LCS) gives unmatched positions $U(y, b)$ and $U(b, y)$; ties advance the candidate side. Define $A_t(y | b) = \text{mean}_{t \in U(y, b)} a_t(y)$. Positions are text tokens, not acoustic alignments; only aggregation is local, while permutation spans the input window. References never locate these positions.

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(a) Frozen models and shared candidate evidence



(b) Alternative selectors

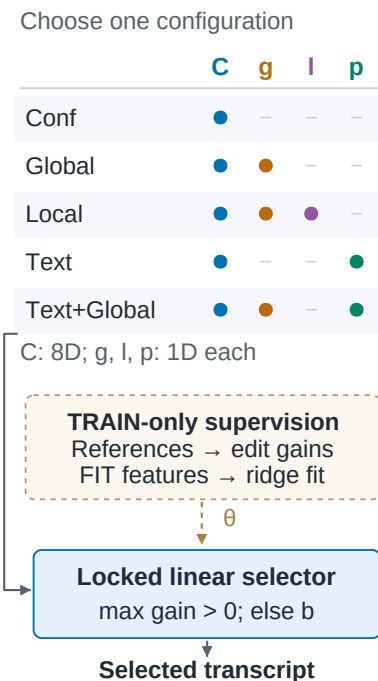


Fig. 1. Frozen reranking architecture. (a) Shared Whisper generation/scoring; only Global, Local and Text+Global use the permuted pass. XLM-R supplies PLL. Synthetic mel blocks and text LCS illustrate permutation and disagreement, not acoustic alignment. (b) Alternative fits: C includes local clean confidence; $g/l/p$ denote global/local contrasts and PLL. Dashed links carry TRAIN coefficients; locks mark frozen models.

3.2. Matched confidence, perturbation and text features

Conf uses eight candidate-minus-beam coordinates: mean clean log probability, candidate probability, log-one-plus body-token length, log-one-plus word length, repeated four-gram proportion (frequency at least three), cap flag, MBR risk and LCS-local clean-confidence difference. Global (9D) appends $A_g(y) - A_g(b)$; Local (10D) also appends $A_l(y \mid b) - A_l(b \mid y)$. Candidate probabilities are a temperature-one softmax of summed clean log scores. MBR minimizes probability-weighted word-edit distance to candidate pseudo-references, each normalized by its own word count (at least one). CD-seq maximizes mean clean log probability plus A_q .

Global and clean-plus-permuted coordinates span the same linear class: $\{c, c - q\}$ spans $\{c, q\}$, also after beam subtraction. Appended controls add either a duplicate clean coordinate (DupClean; no new information) or the beam-relative permuted mean (Perm). FIT RMS and ridge penalties are matched in procedure, not geometry. DupClean halves the effective penalty on its duplicated coordinate. The earlier 10D shuffled control permutes both added columns within FIT language/length strata only.

Frozen XLM-R base [17], trained on multilingual Common-Crawl, masks each non-special body token separately and averages its conditional log probability [7]. Empty text scores zero. Beyond 510 body tokens, fixed centered windows score each target once without truncation. **Text** appends candidate-minus-beam PLL to Conf; **Text+Global** also appends the global coordinate. Both retain Conf’s clean-audio features.

3.3. TRAIN-only fitting and selection

For FIT recording i , candidate y_{ik} and reference r_i , the edit-gain target is $\Delta_{ik} = E(b_i, r_i) - E(y_{ik}, r_i)$, where E counts substitutions, deletions and insertions (S/D/I). With K_i nonbeam candidates and language FIT reference-word total $N_{\ell(i)}$, weights are proportional to $1/(N_{\ell(i)}K_i)$ and normalized to sum to one. Weighted FIT root mean square (RMS) scales features without centering, giving z_{ik} . Each configuration fits

$$\hat{\theta} = \arg \min_{\theta} \sum_{i,k} w_{ik} (\Delta_{ik} - \theta^{\top} z_{ik})^2 + 0.01 \|\theta\|_2^2, \quad (2)$$

with no intercept and matched FIT examples, targets, weights and base coordinates. Beam has a zero vector. Inference chooses the first maximum in bank order for strictly positive gain, otherwise beam. Reference transcripts and reference-derived edit counts are not supplied to the selector at inference. Recording IDs fix the deterministic permutation but are not predictor features.

Exploratory minimum-WER (MWER) fitting uses temperature-one expected edit risk, including beam, with the same penalty. Weights $\propto 1/N_{\ell(i)}$ omit ridge’s candidate-count divisor: objective and weighting both change. Float64 L-BFGS-B selects zero/ridge starts by regularized FIT objective.

4. EXPERIMENTAL PROTOCOL

We evaluate FLEURS [18] and Common Voice v27 (CV27) [19, 20] in Arabic, Persian, Polish, Turkish and Italian. Three Whisper-small checkpoints (seeds 101, 202, 303) use uniform rank-four LoRA

($\alpha=r=4$, no dropout or bias) on query/key/value/output projections in 12 encoder and 24 decoder attention blocks: 144 modules, 884,736 trainable parameters. Six adaptation cycles used 4,656 updates, peak learning rate 2×10^{-4} , 232 warmup updates and effective batch 20; the backbone stayed frozen. Deterministic bfloat16 greedy and width-five beam decoding use known-language prompts, a 444-token cap, no sampling or repetition penalty, and beam length exponent 1.0 (length_penalty=1.0). Reranking freezes ASR and LoRA.

One seed101 TRAIN-FIT fit per selector serves all checkpoints, not three independent fits. Tables 1–2 report test WER; Table 3 uses test samples. DEV/CHECK are labeled. Differences are absolute percentage points, computed before rounding.

FLEURS TRAIN has 13,602 recordings (44.60 h); 1,281 calibrate selectors (1,016 FIT, 265 diagnostic-only CHECK). DEV has 1,731. Calibration selects whole language-specific text-hash groups in SHA order to reach at least 256 recordings/language; a separate hash assigns FIT/CHECK (80/20). Text groups do not imply speaker independence. Each extension locks rules and coefficients before new evaluations; earlier test exposure remains (Sec. 4.1).

WER-v3 fixes Unicode normalization, dotless-i conversion before Turkish lowercasing, apostrophe unification and digit folding, preserving word-internal zero-width non-joiners. WER is $100(S + D + I)/N$ with full reference-word denominator N ; macro WER equally averages five languages. All 3,665 FLEURS TEST recordings (76,164 reference words) and 58,514 CV27 TEST recordings remain. Inputs are capped at 30 s; 22 FLEURS TEST recordings exceed this window and retain full references. No CV27 TEST recording exceeds it. Recording coverage does not imply complete-audio coverage.

4.1. Timeline, uncertainty and cost

The original hypothesis was Local over Global. Original seed101 Local-B5/Conf intervals use 5,000 client-bootstrap draws. Later checkpoint, MWER and Text studies use exposed tests and separate families: 10,000 draws (seed 1729), language-stratified over 11,756 client clusters and paired across methods/checkpoints. Corrections are family-specific; FLEURS is descriptive. Intervals condition on fixed fits/checkpoints, excluding fit uncertainty; crossing zero does not establish equivalence. A separate four-contrast, post-review family gives 98.75% intervals for controls and $\Delta_{\text{cond}} = (W_{\text{Conf}} - W_{\text{Global}}) - (W_{\text{Text}} - W_{\text{Text+Global}})$, with W denoting macro WER. Two disjoint deterministic FIT group subsets (A/B: 513/503 recordings) refit all four ridge configurations with subset-specific scales/weights. All eight fits remain; this descriptive sensitivity check is not independent confirmation.

Seed101 costs use 20 recordings per language/corpus (four per duration quintile, fixed SHA selection). Five systems receive three paired warm runs in rotated order, recomputing candidates and PLL with within-recording encoder sharing. Two A40 GPUs have shared load; loading is separate, and RTF uses processed duration. Intervals resample recordings within corpus/language/quintile after averaging repeats, conditional on observed load.

5. RESULTS AND ANALYSIS

Increment beyond confidence and consensus. Global lowers WER by 0.1410 points on FLEURS and 0.1824 on CV27 relative to Conf (Table 1; Fig. 2(b)). Against the stronger CV27 MBR control, its difference is -0.1093 points; the separate post-hoc 95% client interval

Table 1. Test macro WER (%), mean $\pm$ checkpoint SD; one fit per selector. Oracle6 is reference-dependent. FLEURS retains full references for all 22 recordings exceeding the 30-s input window. MWER and Text are separate post-exposure additions.

Strategy	FLEURS	CV27
<i>Standard controls</i>		
B5	18.877 ± 0.252	36.414 ± 1.186
MBR	18.954 ± 0.241	35.225 ± 0.634
CD-seq	20.793 ± 0.205	39.929 ± 0.622
G	21.128 ± 0.277	39.341 ± 1.494
<i>Matched feature ablations (ridge)</i>		
Conf	18.938 ± 0.285	35.298 ± 0.653
Global	18.797 ± 0.263	35.116 ± 0.640
Local	18.813 ± 0.261	35.125 ± 0.637
<i>Exploratory fitting-objective comparison</i>		
MWER-Conf	18.863 ± 0.260	35.203 ± 0.652
MWER-Global	18.732 ± 0.257	35.000 ± 0.625
<i>Appended frozen-MLM controls</i>		
Text	17.810 ± 0.260	33.912 ± 0.646
Text+Global	17.802 ± 0.238	33.901 ± 0.639
Oracle6 (diagnostic)	16.025 ± 0.196	28.810 ± 0.577

is $[-0.2305, -0.0131]$. This exploratory interval does not inherit the text family’s multiplicity correction.

Local aggregation. Local has slightly higher mean WER than Global on both test corpora. Its CV27 difference is $+0.0092$ points, with an adjusted interval crossing zero (Fig. 2(d)); an incremental benefit is not established. Table 2 reports the language-specific directions.

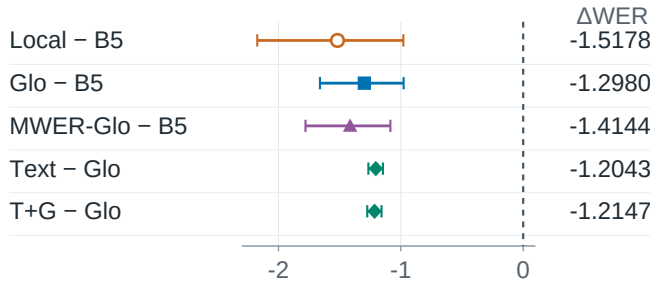
Increment beyond external text. Text improves over Global by 0.9864/1.2043 FLEURS/CV27 points; Text+Global improves over Global by 1.2147 CV27 points. Its -0.0104 difference from Text has interval $[-0.0280, +0.0069]$, establishing neither a further gain nor equivalence (Fig. 2). DEV means are 18.5849/18.5633 for Text/Text+Global, but adding Global worsens seed101 FLEURS DEV/TEST and seed202 CV27. Text’s language-wise reductions relative to Global span 0.67–1.19/0.89–1.55 points; Text+Global–Text stays within ± 0.06 and changes sign (Table 2).

Exploratory fitting comparison. MWER gives lower test point estimates than ridge, but both MWER–ridge contrasts include zero even in ordinary 95% CV27 intervals (Fig. 2(c) shows the adjusted intervals). On DEV, MWER-Global is worse (19.6975 versus 19.6864), while MWER-Conf improves (19.8137 versus 19.8724). Within MWER, Global improves over Conf by 0.1316 FLEURS and 0.2039 CV27 points. Both loss and weighting change.

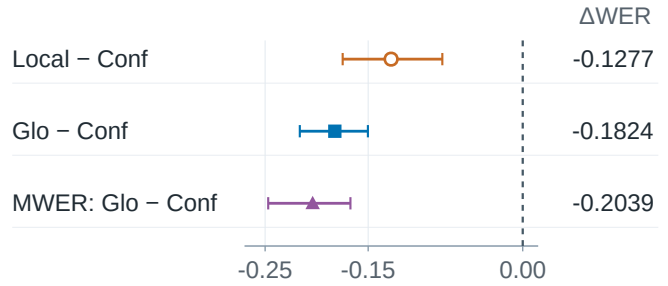
Selection behavior. On seed101 CV27, Global lowers the edit count for 4,272 recordings and raises it for 3,706 relative to B5; 50,536 have equal edit counts, not necessarily identical transcripts. Against Conf, substitutions and insertions decrease but deletions increase. The shuffled-acoustic control gives CHECK WER 15.99 versus Conf’s 15.95 and DEV mean 19.880 versus 19.872.

Parameterization and fit sensitivity. Global minus DupClean is -0.1828 CV27 points [98.75% CI: $-0.2189, -0.1497$]; this duplicate-coordinate control does not recover Global’s gain. Perm minus Conf is -0.1834 [$-0.2193, -0.1497$]; Global minus Perm is $+0.0011$ [$-0.0008, +0.0029$]. Thus a contrast-specific advantage is not established. The paired Δ_{cond} is $+0.1720$ [$+0.1362, +0.2090$]; the incremental WER reduction is smaller with Text for

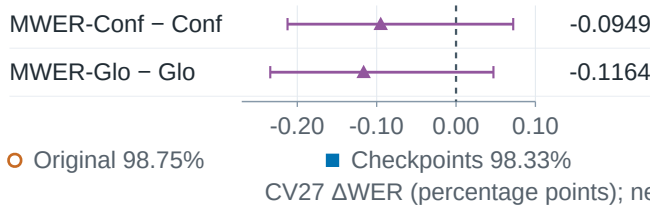
(a) Shared-bank comparisons



(b) Audio beyond confidence



(c) Alternative fitting objective



(d) Near-zero increments · expanded axis

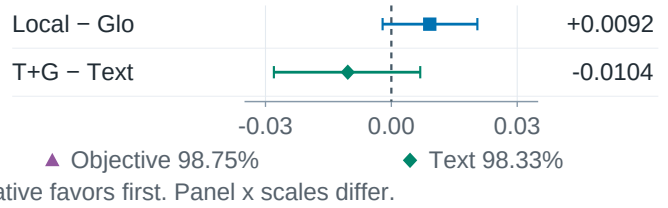


Fig. 2. Paired CV27 WER differences; family-specific client intervals condition on fixed fits. Original: seed101, 98.75%; Checkpoints/Text: three-checkpoint mean, 98.33% (rounded from 98.333333%); Objective: the same mean, 98.75%. Glo=Global, T+G=Text+Global. MWER: Glo-Conf compares MWER fits. Scales differ; (d) enlarges zero. All 12 contrasts appear once.

Table 2. Three-checkpoint language WER (%). Δ_L =Local-Global, Δ_T =Text+Global-Text; positive is worse. Bold denotes the lowest point estimate, not significance. Glo/T+G=Global/Text+Global.

Lang.	Absolute WER				Differences	
	Conf	Glo	Text	T+G	Δ_L	Δ_T
<i>FLEURS</i>						
ar	26.618	26.510	25.315	25.278	+0.012	-0.037
fa	32.109	32.049	30.963	30.947	+0.003	-0.017
pl	15.162	14.944	14.036	14.019	+0.009	-0.016
tr	13.335	13.153	12.078	12.093	+0.057	+0.015
it	7.465	7.328	6.660	6.671	-0.002	+0.011
<i>CV27</i>						
ar	53.715	53.508	52.621	52.564	-0.003	-0.056
fa	67.382	67.326	65.950	65.955	+0.004	+0.005
pl	17.257	16.938	15.390	15.385	-0.006	-0.005
tr	22.766	22.548	21.493	21.478	+0.036	-0.014
it	15.373	15.260	14.105	14.124	+0.015	+0.019

fixed fits. Coefficients are refit across configurations; this pipeline comparison tests neither causal interaction nor information redundancy.

Subset A/B fits lower CV27 WER for Global relative to Conf by 0.2179/0.3379 points. Text+Global-Text is -0.0231 for A but $+0.8816$ for B; on FLEURS it is $-0.0276/ -0.0132$. This shows sensitivity to the FIT subset and its scales and weights; main intervals condition on the original fits. Neither subset replaces the original model.

Acquisition cost. Under shared load, summed warm-time ratios for Global/Conf are 1.036 on FLEURS and 1.097 on CV27; Text+Global/Text ratios are 1.044 and 1.085. Pooled ratios are 1.054 [descriptive 95% sampling interval: 1.049, 1.059] and 1.056

Table 3. Warm-inference cost, seed101: 100 recordings per corpus, three warm runs per system, two NVIDIA A40 GPUs. RTF uses processed duration; med/p95 is sentence latency (seconds). Candidates and scores are recomputed after model loading.

Pipeline	FLEURS		CV27	
	RTF	med/p95	RTF	med/p95
B5	0.122	1.35/3.01	0.121	0.56/1.20
Conf	0.235	2.63/5.69	0.236	1.08/2.37
Global	0.244	2.74/5.89	0.259	1.21/2.38
Text	0.255	2.79/6.39	0.256	1.20/2.40
Text+Global	0.266	2.99/6.59	0.277	1.31/2.60

[1.050, 1.061]. Conf/B5 is 1.930 [1.921, 1.940]: the small marginal Global cost excludes the common reranking overhead. These intervals do not cover time-varying contention. XLM-R raises peak allocation from 833 to 2,105 MiB and adds a 1.13 GB snapshot. External GPU processes appeared at 1,672/3,000 timing boundaries on one card; different recording subsets preclude between-card speed comparisons.

6. LIMITATIONS AND CONCLUSION

Permutation scores modestly improve confidence-based selection, without established benefit beyond external text or a contrast-specific advantage. Evidence covers one backbone, observed tests and 30-second inputs, not full-audio recognition. Fit sensitivity, additional model resources and shared-load timings limit deployment conclusions.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This study reanalysed existing FLEURS and CV27 releases under their published terms, without participant recruitment, new recordings or speaker identification. The authors are responsible for substantive arguments, verification and final approval.

8. FUNDING ACKNOWLEDGMENTS

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